

Low-Complexity 2-D Digital FIR Filters Using Polyphase Decomposition and Farrow Structure

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Abstract—This paper proposes a novel realization technique for quadrantly symmetric 2-D finite impulse response filters with a guaranteed reduction in the hardware complexity. Here, the concept of Farrow structure-based interpolation filter design using the polyphase decomposition of the 1-D filter transfer function is effectively utilized in the 2-D domain. The proposed 2-D filter makes use of row-wise polyphase decomposition of the 2-D transfer function or frequency response, followed by the polynomial approximation of the individual polyphase coefficients resulting in Farrow structures corresponding to each row filter. The final coefficients are implemented by varying the delay values in all the Farrow structures, followed by the interpolation of the coefficients obtained from each delay value, which in turn forms the rows in the 2-D kernel. The major highlight of the proposed method is the highly reduced implementation complexity in terms of the number of multipliers and adders, with a low normalized root-mean-square error. Design examples of the circularly symmetric and fan-type filters have been considered to show the efficiency of the approach. The results show a drastic reduction in the implementation complexity of the 2-D filters of upto 20%, with significantly low normalized root-mean-square error lesser than 0.5%.

Index Terms—Two dimensional (2-D) FIR filter, Farrow structure, P_2 transformation, circularly symmetric filters, fan type filters.

I. INTRODUCTION

THE design and implementation of 2-D filters is a crucial area, widely explored by researchers due to the wide variety of applications they offer. The application areas include pattern recognition systems, computer image processing, seismic signal processing, biomedical systems, and genomic signal processing. Since many critical applications such as real-time medical image analysis, require fast 2-D filtering operations with filter kernels of large size [1], [2], effective, fast, low complexity and low power implementation approaches for 2D filters, become essential.

Among the different design approaches for 2D filters [3]–[21], the transformation based approach is a

widely used design method [14], [16]–[21] for circular and fan-type filters. In the transformation-based method, a suitable 1D filter is designed first, followed by mapping the 1D filter coefficients to 2D domain using specific transformation kernels. McClellan transformation [14] was the first transformation introduced for designing circular 2D filters from their 1D counterparts. In [19], two new multiplierless transformations were introduced, which was a remarkable achievement over McClellan transformation, towards the aim of designing circular 2D filters of improved circularity. Later, with an aim of designing 2D filters of lower implementation complexity and improved circularity, a new P_2 transformaton was proposed in [21]. The McClellan transformation based design approach was also extended for the design of fan filters in [14], [16], and [17], where the transformation coefficients were deduced using different optimization algorithms by maintaining a minimum contour deviation error. Bindima *et al.* [22] have deduced the transformation coefficients for fan-type filters using multi-objective artificial bee colony optimization by ensuring the least contour deviation error.

The 2D filter designed using the transformation method has an efficient implementation structure based on Chebyshev recursion [23] formula, which consists of cascaded 2D transformation kernels $F(z_1, z_2)$, where $F(z_1, z_2)$ represents the 2D transformation function corresponding to the spectral transformation $F(\omega_1, \omega_2)$ used. But the Chebyshev structure based implementation in the z domain has a major disadvantage of increased delay due to the use of cascaded 2D kernels $F(z_1, z_2)$, which in turn act as a limitation to the high speed filtering. The implementation of a 2D filter of size $(2N+1) \times (2N+1)$ requires a cascade of N number of similar $F(z_1, z_2)$ blocks. This, in turn, introduces an overall delay of $z_1^{-N} z_2^{-N}$ for the transformation-based realizations. Moreover, the realization of the transformations such as P_2 that provide better contour circularity requires fractional delay filters to realize $z^{-\frac{1}{2}}$. An alternative solution to overcome the delays involved in the 2D filter realization is to use the Chebyshev structure for frequency domain-based implementation of the 2D filter as in [21]. But the realization of the 2D filter in the frequency domain requires larger number of multipliers and thus increases the implementation complexity.

Apart from transformation-based methods, different optimization based methods were also introduced for the design of 2D filters, where the entire $(2N+1) \times (2N+1)$ number

Manuscript received September 6, 2018; revised November 25, 2018; accepted December 6, 2018. This paper was recommended by Associate Editor G. Jovanovic Dolecek. (Corresponding author: T. Bindima.)

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Digital Object Identifier 10.1109/TCSI.2018.2889260

of 2D filter coefficients for specific contours, are obtained by minimizing the contour deviation error [5], [9], [15], [24]. The realization of these 2D filters relies upon the direct form implementation of the 2D transfer function. Thus, the realization of a 2D filter of size equal to $(2N + 1) \times (2N + 1)$ requires $2N$ delays both in the z_1 and z_2 space. On the other hand, the frequency domain realization of the filters increases the implementation complexity. Recently, a computationally efficient 2D filter design approach using sampling-kernel based interpolation and frequency transformation is proposed for the design of sharp 2D FIR filters [13], which focusses on reducing the implementation complexity of the 2D filters. The major drawback of this approach is the design complexity in yielding the optimal lengths of the different sub-filters.

This paper proposes a novel implementation structure for 2D filters of reduced hardware complexity in frequency domain. The proposed implementation structure makes use of the theory of the design of interpolation filters using 1D Farrow structure [25]. The proposed implementation structure realizes the polyphase components corresponding to each row in the filter kernel using the Farrow structure, and the overall 2D filter is realized using a set of parallel Farrow structures, all tunable with the same fixed set of fractional delays. The coefficients obtained from each Farrow structure using the different fractional delays are suitably interpolated to realize the individual rows in the 2D kernel. The total delay involved in the realization of the individual row filters reduces to K , where K represents the order of each polyphase subfilter. The proposed low complexity implementation of the 2D filter has been demonstrated for circular 2D filters and fan-type filters, and can be extended to other types such as elliptical and diamond filters. The quadrantal symmetry [26] of the 2D circular and fan-type filter kernels, which are designed using the existing transformation based approaches, ensure the filter coefficients $h(n_1, n_2)$ to satisfy the condition $h(n_1, n_2) = h(N - n_1, n_2) = h(n_1, N - n_2)$ [26] for a 2D filter of order equal to $N \times N$. This can also be effectively utilized in reducing the hardware complexity. The proposed method offers effective hardware implementation and better complexity reduction than the state of art methods.

The paper is organized as follows. Section II gives a brief insight into the concept of interpolation filters based on Farrow structure and the transformations used in this work for the design of 2D filters. Section III explains the proposed methodology for the design and implementation of low complexity 2D filters with a detailed mathematical analysis. The proposed design methodology is demonstrated using design examples in section IV, while the results and discussions are considered in section V, and section VI concludes the paper.

II. PRELIMINARIES

A. Farrow Structure Based Interpolation Filters

Farrow structure based interpolation filter design approach [25], [27] has been reported to be an efficient low complexity method for the implementation of higher order 1D FIR interpolation filters. The implementation scheme makes use of the polyphase decomposition of the filter transfer function. The individual coefficients of all the polyphase components

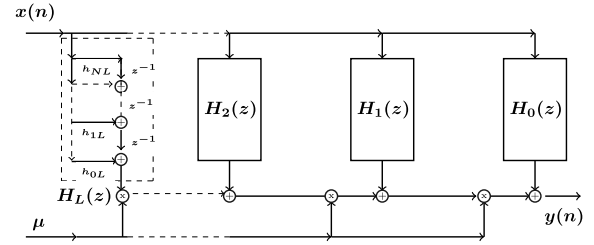


Fig. 1. Farrow structure with fixed sub-filters $H_k(z)$ and the delay parameter (μ) [29].

are taken together and approximated using an L^{th} order polynomial in a set of delay values μ . This polynomial approximation is closely related to the concept of Taylor polynomial approximation of an infinite sequence using a finite order polynomial. This polynomial approximation results in a set of polynomials equal to the number of coefficients in each polyphase term, which forms the basic Farrow structure [28] shown in Fig. 1. The coefficients of the different polynomials form the different sub-filter coefficients of the Farrow structure. The different sub-filters $H_k(z)$ are in fact linear phase FIR filters and are implemented in transposed form as in Fig. 1. Thus, the individual coefficients of the filter are realized from the Farrow structure by changing the delay values, followed by interpolation. As in Taylor polynomial approximation, the bound value of μ should be chosen in the range $|\mu| \leq 1$, in order to maintain the error in approximation to a minimum.

B. 2D Filter Design Using Transformation Method

Transformation based design of 2D filters is a frequency domain design approach. The design starts with a suitable 1D FIR filter of order $2N$ with frequency response $H(\Omega)$, and a mapping function $\cos \Omega \rightarrow F(\omega_1, \omega_2)$ is applied in $H(\Omega)$ to form the 2D frequency response $H(\omega_1, \omega_2)$ as given in Eqs 1-2.

$$\begin{aligned} H(\Omega) &= \sum_{n=-N}^N h_n e^{-j\Omega n} \\ &= h_0 + 2 \sum_{n=1}^N h_n \cos n\Omega = h_0 + 2 \sum_{n=1}^N h_n T_n(\cos \Omega) \end{aligned} \quad (1)$$

$T_n(x)$ represents the n^{th} order Chebyshev polynomial defined by the recursive equation $T_n(x) = 2xT_{n-1}(x) - T_{n-2}(x)$, where the initial values are given by $T_0(x) = 1$, $T_1(x) = x$.

$$H(\omega_1, \omega_2) = h_0 + 2 \sum_{n=1}^N h_n T_n(F(\omega_1, \omega_2)) \quad (2)$$

In the above equation, $F(\omega_1, \omega_2)$ is a 2D mapping function [30], which varies according to the transformation used. Thus, the transformation based approach is a flexible design approach, where the 2D filter coefficients in the frequency domain are directly obtained from their 1D impulse response coefficients by applying a frequency transformation. McClellan transformation is the first reported transformation, which has been used for the design of different 2D filters

TABLE I
DESIGN VARIABLES FOR THE TRANSFORMATION-BASED
DESIGN OF QUADRANTAL FAN FILTERS [22]

Slope θ	Parameters for the design of fan filters $\frac{\omega_c}{\pi}$	t_{00}	t_{01}
10°	0.9027	-0.3119	-0.6657
15°	0.7778	-0.3004	-0.6495
20°	0.7988	-0.2626	-0.6428
25°	0.7501	-0.2449	-0.6126
30°	0.6973	-0.2106	-0.58488
35°	0.6376	-0.1578	-0.5569
40°	0.5726	-0.09086	-0.52409

such as circular, fan type and elliptical filters [17], [31]–[35]. For the design of circularly symmetric filters, with an aim to improve the contour circularity offered by McClellan transformation several frequency transformations have been introduced in the literature. These include generalized McClellan, T1 [19], T2 [19], P1 and P2 [21] transformations. Out of these T1 and P2 transformations provide good contour circularity. In this paper, we concentrate on the design and implementation of circular and fan-type filters.

1) *Circular 2D Filters*: Among the different transformations for circular 2D filters, T1 produce 2D filters with perfect contour circularity but of slightly increased size. P2 transformation produces smaller sized filters with comparatively better contour circularity than the remaining transformations. Thus, P2 transformation has been used in this work for demonstration. The mapping function $F(\omega_1, \omega_2)$ represented as $F_{P2}(\omega_1, \omega_2)$ for P2 transformation is given by (3).

$$F_{P2}(\omega_1, \omega_2) = 2 \left[\cos \frac{\omega_1}{2} \cos \frac{\omega_2}{2} \right]^{2k} \left[0.25 \left(\cos \frac{\omega_1}{2} + \cos \frac{\omega_2}{2} \right)^2 - 1 \right] \quad (3)$$

2) *Fan-Type 2D Filters*: Direct design of quadrantly symmetric fan-type 2D filter with arbitrary slope using McClellan transformation is a well accepted design method with an established implementation structure. The fan filters of the desired slope angle (θ) and PB region of interest (ROI) are defined by the cut-off line $\omega_2 = k\omega_1$ for $k = \tan \theta$ as in Fig. 2. These filters are designed using McClellan transformation given by (4) [14], [34]

$$F(\omega_1, \omega_2) = t_{00} + t_{10} \cos \omega_1 + t_{01} \cos \omega_2 + t_{11} \cos \omega_1 \cos \omega_2 \quad (4)$$

with proper selection of the transformation coefficients t_{ij} ($i, j \in [0, 1]$), so that the contour deviation error is maintained to have the least value. McClellan transformation based design approach transforms a 1D filter of suitable cut-off frequency directly to 2D. The cut-off frequency of the 1D filter and the transformation coefficients used for mapping the 1D filters to 2D depend on the slope angles, which are detailed in Table. I [22].

III. PROPOSED METHODOLOGY FOR LOW COMPLEXITY 2D FILTER

The 2D filter coefficients for any given specifications are obtained using any of the existing design approaches [13], [19], [21], [24]. On obtaining the filter coefficients, the filters designed using the optimization based methods as in [24] are

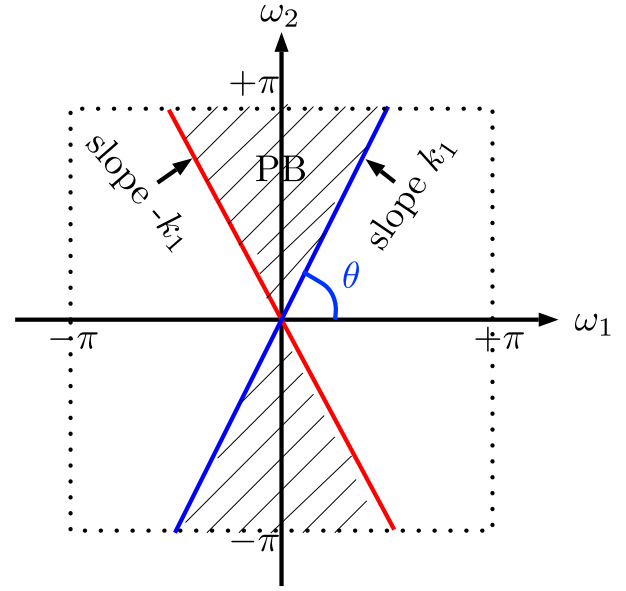


Fig. 2. Contour of an ideal fan-type filter [22].

realized using direct form structure and those designed using the transformation based methods [19], [21] are realized using the Chebyshev structure either in frequency domain or time domain. Eventhough the time domain realization of the 2D filter based on Chebyshev recursion formula as in [19], [21], and [22] provides a low complexity implementation structure, the delay involved in the 2D filtering operation using this structure is very high. A suitable solution is to make use of the frequency domain realization using the Chebyshev structure as detailed in [21], where delay can be reduced due to direct multiplication of the incoming 2D signal with the coefficients of the 2D transformation kernel and the 1D filter coefficients. But, on the other hand, the implementation complexity of this approach is very high. A trade-off between the implementation complexity and delay is to be maintained, which is ensured by the proposed low complexity implementation structure in frequency domain. The major highlight of the proposed technique is that it is compatible for time-domain as well as frequency-domain based realization. The proposed method utilizes the concept of polynomial approximation based interpolation filter design for realizing the individual row filters in the 2D frequency response. In general, the frequency response of a 2D filter is given by

$$H(\omega_1, \omega_2) = \sum_{n_1=0}^{N_1} \sum_{n_2=0}^{N_2} h(n_1, n_2) \exp(-jn_1\omega_1) \exp(-jn_2\omega_2) \quad (5)$$

where $h(n_1, n_2)$ represents the 2D filter coefficients of the filter of size $(N_1 + 1) \times (N_2 + 1)$. The proposed filter structure is based on the 1D polyphase decomposition of each row of the 2D filter after the coefficients in the filter kernel are derived using any of the existing design methods. The filter coefficients of each row in the 2D filter kernel, obtained by transformation method, corresponds to a linear phase 1D FIR row filter.

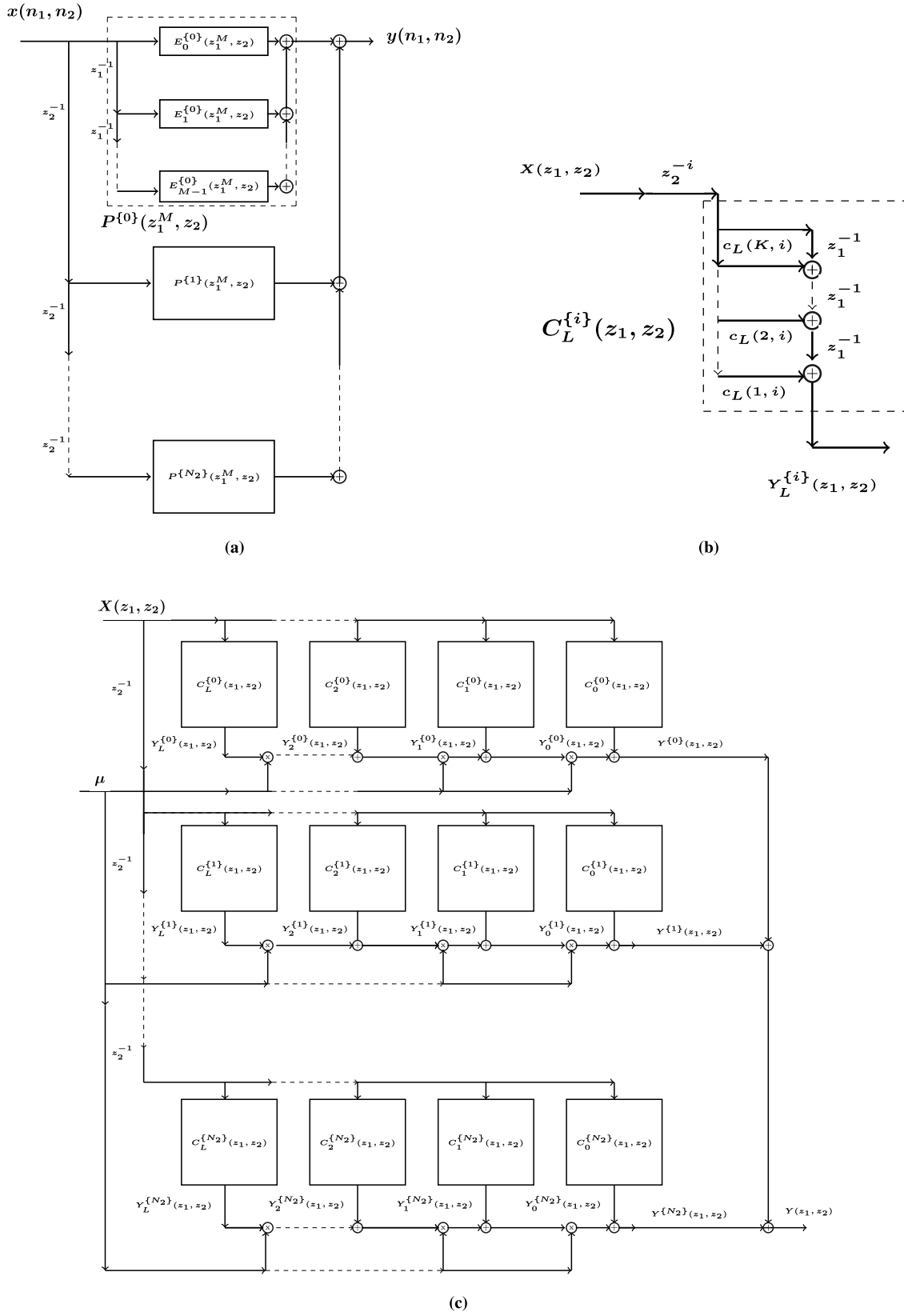


Fig. 3. Proposed 2D filter (a) 2D filters with 1D polyphase decomposition (b) Realization of individual sub-filters $C_k^{i\{i\}}(z_1, z_2)$ in the proposed structure (c) Proposed 2D filter using Farrow structure based interpolation filter.

A. 1D Polyphase Decomposition

The proposed approach starts with the polyphase decomposition of the transfer function corresponding to the individual

rows of the 2D filter kernel as shown in Fig. 3a. Here, first we consider the case of 2D filters of order $N_1 \times N_2$ with even N_1 . The 2D transfer function is expressed using M^{th}

order polyphase decomposition of each of the individual i^{th} row in the 2D filter kernel (for $i = 0, 1..N_2$) as

$$\begin{aligned} H(z_1, z_2) &= \sum_{i=0}^{N_2} z_2^{-i} P^{(i)}(z_1, i) \\ &= \sum_{i=0}^{N_2} z_2^{-i} \left(\sum_{m=0}^{M-1} z_1^{-m} E_m(z_1^M, i) \right) \end{aligned} \quad (6)$$

The polyphase components $E_m(z_1, i)$ corresponding to each of the i^{th} row in (6) can be represented in a generalized form as $E_m^{(i)}(z_1, z_2)$. In the case of even order row filters, the interpolation order M for polyphase decomposition is to be chosen such that $N_1 = MN_0$, for any integer N_0 . The polyphase components of each of the i^{th} row filter can be expressed as

$$E_m^{(i)}(z_1, z_2) = \sum_{n_1=0}^{\infty} h(m + n_1 M, i) z_1^{-n_1} \quad (7)$$

For the case of a 2D low pass filter (LPF) having quadrantly symmetric response, the individual row filters should ideally represent an ideal LPF, so that $H(z_1, i) \approx z_2^{-i} z_1^{-\frac{N_1}{2}}$. This implies that the different polyphase components ($E_m^{(i)}(z_1, z_2)$) of each of the i^{th} row filter closely approximates an all pass function with a delay equal to $\frac{N_1}{2M} - \frac{m}{M}$ along the z_1 direction ($m=0,1,..M-1$) in the pass band (PB) region, so that

$$E_m^{(i)}(z_1, z_2) \approx z_2^{-i} z_1^{-\left(\frac{N_0}{2} - \frac{m}{M}\right)}, \quad \omega \in \left[-\frac{\omega_p}{\pi}, \frac{\omega_p}{\pi}\right] \quad (8)$$

Hence, in the case of even N_1 , the zeroth polyphase component $E_0^{(i)}(z_1, z_2)$ of each row approximates a fixed integer delay $z_1^{-\frac{N_0}{2}}$, while the remaining polyphase terms $E_m^{(i)}(z_1, z_2)$ (for $m = 1, 2, ..M-1$) closely approximate a fractional delay of $z_1^{-\left(\frac{N_0}{2} - \frac{m}{M}\right)}$. The polyphase components of each i^{th} row filter approximating fractional delays can be expressed in matrix form as

$$\begin{aligned} &\begin{bmatrix} E_1^{(i)}(z_1, z_2) \\ E_2^{(i)}(z_1, z_2) \\ \vdots \\ E_{M-1}^{(i)}(z_1, z_2) \end{bmatrix} \\ &= \begin{bmatrix} h(1, i) & h(M+1, i) & \dots & h((K-1)M+1, i) \\ h(2, i) & h(M+2, i) & \dots & h((K-1)M+2, i) \\ \dots & \dots & \dots & \dots \\ h(M-1, i) & h(2M-1, i) & \dots & h(KM-1, i) \end{bmatrix} \\ &\quad \times \begin{bmatrix} 1 \\ z_1^{-1} \\ \vdots \\ z_1^{-(K-1)} \end{bmatrix} \end{aligned} \quad (9)$$

where $K = N_0$ and the polyphase coefficient matrix corresponding to the i^{th} row of the 2D filter is given

$$P^{(i)} = \begin{bmatrix} h(1, i) & h(M+1, i) & \dots & h((K-1)M+1, i) \\ h(2, i) & h(M+2, i) & \dots & h((K-1)M+2, i) \\ \dots & \dots & \dots & \dots \\ h(M-1, i) & h(2M-1, i) & \dots & h(KM-1, i) \end{bmatrix} \quad (10)$$

B. Polynomial Approximation of the Polyphase Coefficients

The low complexity 2D filter is derived by polyphase decomposition of the individual row filters of the 2D kernel and by approximation of each of the polyphase coefficients using a polynomial in terms of a set of fractional delays. Filter coefficient implementation using polynomial approximation in terms of the fractional delays is based on the concept of the design of 1D interpolation filters using Farrow structure. The polynomial approximation in terms of the fractional delay, works effectively for the coefficients of the polyphase components $E_m^{(i)}(z_1, z_2)$ for $m = 1, 2, ..M-1$, since they approximate a fractional delay equal to m/M . The coefficients in each m^{th} row of the polyphase coefficient matrix $P^{(i)}$ correspond to the coefficients of the polyphase term $E_m^{(i)}(z_1, z_2)$. The polyphase decomposition of the row filters of $H(z_1, z_2)$ of order N_1 using an interpolation order of M , results in M number of polyphase terms corresponding to each row of $H(z_1, z_2)$. The number of coefficients in each polyphase term is N_0 , except for the zeroth polyphase term, where it is $N_0 + 1$. But, since the zeroth polyphase component approximates an integer delay, it is excluded from polynomial approximation. Hence, $P^{(i)}$ is of dimension $(M-1) \times K$. The coefficients in any j^{th} ($j = 1, 2, ..K$) column of $P^{(i)}$ represents the j^{th} coefficient from all the $M-1$ polyphase terms. For $j = 1$, the coefficient set $h_1 = [h(1, i) \ h(2, i) \ \dots \ h(M-1, i)]^T$ corresponding to the first column of $P^{(i)}$ represents the set of first coefficients from the $M-1$ polyphase terms $E_m^{(i)}(z_1, z_2)$ for $m = 1, 2, ..M-1$. These coefficients in turn, correspond to the coefficients of different filters approximating varying fractional delays equal to m/M . Each coefficient set h_j consisting of $M-1$ coefficients is polynomial approximated to an L^{th} order polynomial in terms of the delay parameter μ , where μ takes a set of $M-1$ values equal to m/M for $m = 1, 2, ..M-1$. Hence, the approximated L^{th} order polynomial takes the form $h_j = c_L(j, i)\mu^L + c_{L-1}(j, i)\mu^{L-1} + \dots + c_1(j, i)\mu + c_0(j, i)$, so that the polynomial evaluation with any $\mu = m/M$ reconstructs the m^{th} coefficient in h_j . The polynomial approximation is done for all h_j ($j = 1, 2, ..K$) so that (9) can be rewritten as

$$\begin{aligned} &\begin{bmatrix} E_1^{(i)}(z_1, z_2) \\ E_2^{(i)}(z_1, z_2) \\ \vdots \\ E_{M-1}^{(i)}(z_1, z_2) \end{bmatrix} = \begin{bmatrix} \mu_1^L & \mu_1^{L-1} & \dots & \mu_1 & 1 \\ \mu_2^L & \mu_2^{L-1} & \dots & \mu_2 & 1 \\ \vdots & \vdots & \dots & \vdots & \vdots \\ \mu_{M-1}^L & \mu_{M-1}^{L-1} & \dots & \mu_{M-1} & 1 \end{bmatrix} \\ &\quad \times \begin{bmatrix} c_L(1, i) & c_L(2, i) & \dots & c_L(K, i) \\ c_{L-1}(1, i) & c_{L-1}(2, i) & \dots & c_{L-1}(K, i) \\ \vdots & \vdots & \dots & \vdots \\ c_0(1, i) & c_0(2, i) & \dots & c_0(K, i) \end{bmatrix} \begin{bmatrix} 1 \\ z_1^{-1} \\ \vdots \\ z_1^{-(K-1)} \end{bmatrix} \end{aligned} \quad (11)$$

or

$$E = \mu^T C z_1^T \quad (12)$$

which can be diagrammatically demonstrated using Fig. 3a-3c. The polyphase decomposition of the individual row filters of the 2D kernel is shown in Fig. 3a. The overall structure of the proposed low complexity 2D filter of size $N_1 \times N_2$ is shown in Fig. 3c and the realization of the individual sub-filters in the proposed 2D structure is shown in Fig. 3b. The proposed structure consists of N_2 levels, where each level corresponds to each row filter. In each level, the row filter is implemented using polynomial approximation. The polynomial coefficients of each μ^k for $k = 0, 1, \dots, L$, together form the sub-filters in the proposed 2D structure. Hence, each level consists of $L + 1$ number of linear phase sub-filters which are 1D FIR filters. In each level, the filter coefficients reconstructed from the Farrow structure for each value of μ are suitably interpolated by an order equal to M to reconstruct the coefficients of the row filter. For the case of 2D filters of order $N_1 \times N_2$ with odd N_1 , the interpolation order M for polyphase decomposition is chosen in such a way that $N_1 + 1 = N_0 M$. In this case, the zeroth polyphase term are also considered for polynomial approximation, since these terms also correspond to fractional delay filters.

IV. DESIGN EXAMPLES

This section demonstrates the proposed methodology for low complexity 2D circular filters and fan-type filters using different design examples. 2D circular LPFs designed using P_2 transformation [21] are considered in design examples I and II. The effectiveness of the proposed approach for fan-type filters is also demonstrated in example III, after finding the 2D filter coefficients using McClellan transformation. All these transformations produce quadrantly symmetric impulse response.

A. Example I

The design example of the LPF in [21], with the following specifications has been considered here.

$$H(\omega_1, \omega_2) = \begin{cases} 1 + \delta_p, & 0 \leq \sqrt{\omega_1^2 + \omega_2^2} \leq 0.75\pi \\ \delta_s, & 0.9\pi \leq \sqrt{\omega_1^2 + \omega_2^2} \leq \pi \end{cases} \quad (13)$$

where, $\delta_p = \delta_s = 0.01$. The respective 2D PB and SB radial edge frequencies $\omega_{pr} = 0.75\pi$ and $\omega_{sr} = 0.9\pi$ are reverse mapped using the relation in (14) for P_2 transformation, to yield the corresponding 1D edge frequencies Ω_p and Ω_s .

$$\Omega = \cos^{-1}[(2[\cos(\omega_1/2)\cos(\omega_2/2)]^{2k} - 1)] \quad (14)$$

Here k is chosen as 1. To find the PB edge frequency Ω_p , the 2D frequencies are taken as $\omega_1 = \omega_{pr} \cos \theta$ and $\omega_2 = \omega_{pr} \sin \theta$, where the angle θ varies from 0 to 2π . Similarly, for $\Omega = \Omega_s$, $\omega_1 = \omega_{sr} \cos \theta$ and $\omega_2 = \omega_{sr} \sin \theta$. Thus, each 2D radial frequency when reverse mapped results in a set of 1D edge frequencies in the range $[\Omega_{p,min}, \Omega_{p,max}]$ and $[\Omega_{s,min}, \Omega_{s,max}]$, respectively for the PB and SB. Thus,

the reverse mapping of the 2D edge frequencies result in the 1D cut-off edges equal to $[\Omega_p, \Omega_s] = [0.7331\pi, 0.8621\pi]$. These specifications result in a 1D filter of order equal to 30, using Parks-McClellan (PM) algorithm. The 1D filter when transformed to 2D using P_2 mapping, results in a 2D filter of size equal to 61×61 . The polyphase decomposition of the row filters is performed using an interpolation order of $M = 15$ resulting in $N_0 = 4$. For each i^{th} row of the 2D kernel, polyphase decomposition results in 15 polyphase terms $E_m^{(i)}(z)$ ($m = 0, 1, \dots, 14$). The polyphase terms $E_1^{(i)}(z), E_2^{(i)}(z), \dots, E_{14}^{(i)}(z)$ consist of four coefficients. While the zeroth polyphase term $E_0^{(i)}(z)$ consists of five coefficients. The magnitude responses and the phase responses of different polyphase terms of different row filters plotted using filter visualization tool is shown in Fig. 4. It is observed that, the zeroth polyphase term offers a constant phase delay of $\frac{N_0}{2} = 2$ and the remaining polyphase terms offer a fractional phase delay approximately equal to $\frac{N_0}{2} - \frac{m}{M}$ in the PB of the filter. Due to the quadrantal symmetry of the 2D filter, only the filter coefficients of half the total number of rows need to be considered for polyphase decomposition and polynomial approximation. Hence, 31 row filters are to be considered for the same. Only the polyphase subfilters that offer fractional phase delays are considered for polynomial approximation. Thus, the polyphase decomposition of each of the 31 rows using an interpolation order of 15, results in a polyphase coefficient matrix $\mathbf{P}^{(i)}$ of size equal to 14×4 for $i = 1, 2, \dots, 31$. The rows of $\mathbf{P}^{(i)}$ represent the polyphase subfilters with fractional phase delay. Each column correspond to the corresponding coefficients of the different polyphase subfilters. For eg, the first column of $\mathbf{P}^{(i)}$ represent the first coefficients from $E_1^{(i)}(z), E_2^{(i)}(z), \dots, E_{14}^{(i)}(z)$ obtained by polyphase decomposition of the i^{th} row in the transfer function of the 2D filter. The individual columns of each $\mathbf{P}^{(i)}$ is polynomial approximated to an eighth order polynomial in the variable delay parameter μ . Since each column includes 14 coefficients, a set of 14 fractional delay values (μ), equally distributed in the range $[-0.5, 0.5]$ is used. The optimal polynomial order is chosen in such a way that the maximum deviation error between the original 2D filter response and the response of the 2D filter reconstructed from the proposed structure falls within a very small limit. An eighth order polynomial approximation is used in this case to form the Farrow structure. For each row, the eighth order polynomial approximation results in a set of nine Farrow subfilters each of size equal to four, where the subfilters exhibit coefficient symmetry. The direct form implementation of the 2D filter of kernel size equal to 61×61 involves 961 coefficient multipliers, and 3600 adders. But the proposed approach reduces the implementation complexity to 931 multipliers and 1230 adders. The error measures used to find the optimum polynomial order are mean square error (MSE) and normalized root mean square error (NRMSE). If $H(\omega_1, \omega_2)$ represents the frequency response of the original 2D filter and if $Y(\omega_1, \omega_2)$ represents the frequency response of the 2D filter obtained from the proposed approach, then the MSE and NRMSE between $H(\omega_1, \omega_2)$ and $Y(\omega_1, \omega_2)$ are given by (15) and (16), respectively. The proposed method for

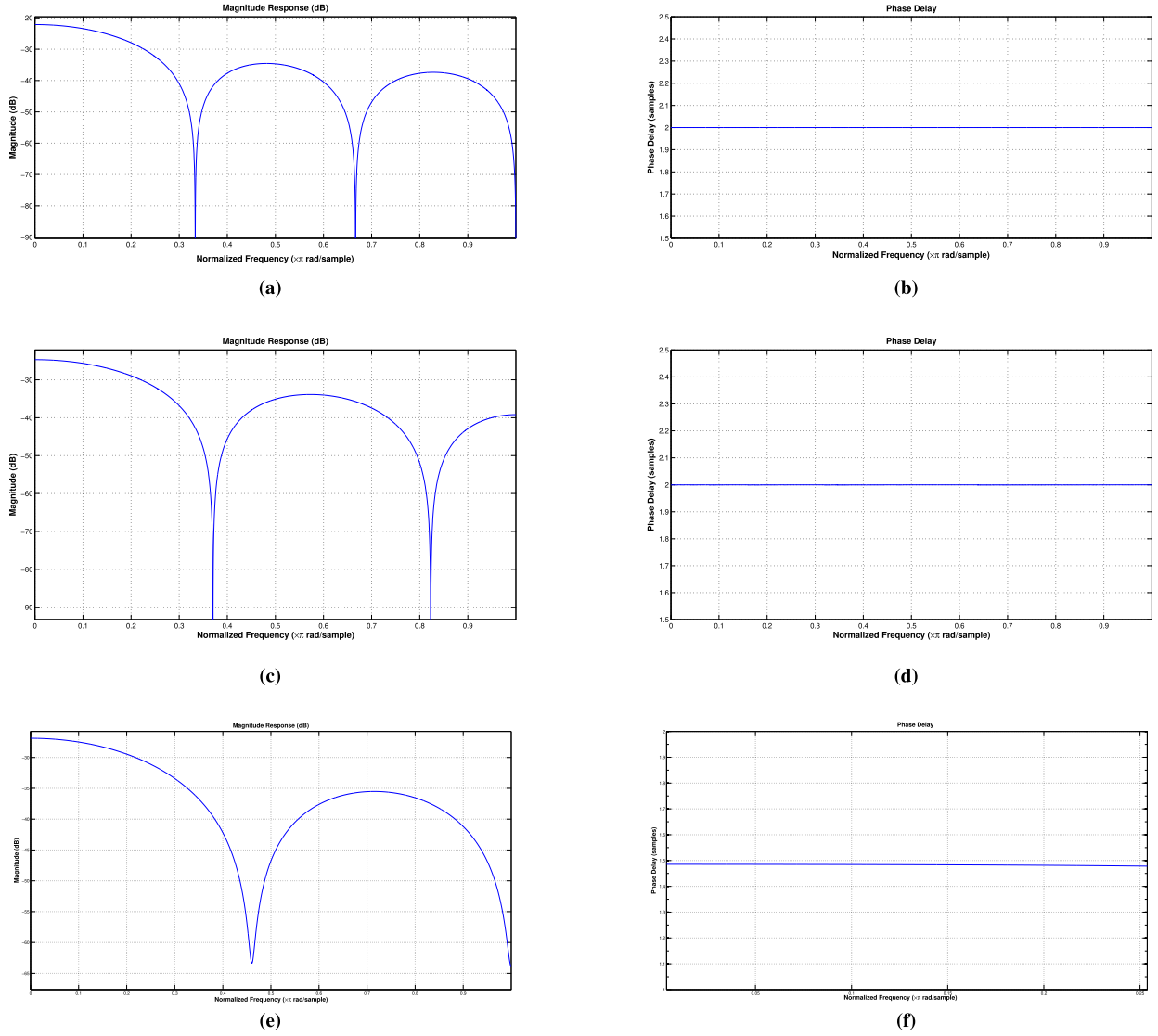


Fig. 4. Magnitude and phase response plots obtained for different polyphase terms corresponding to different rows in the 2D kernel of design example 1. (a) Magnitude response of $E_0^{(1)}(z_1, z_2)$, (b) Phase response of $E_0^{(1)}(z_1, z_2)$, (c) Magnitude response of $E_7^{(1)}(z_1, z_2)$, (d) Phase response of $E_7^{(1)}(z_1, z_2)$, (e) Magnitude response of $E_7^{(2)}(z_1, z_2)$, (f) Phase response of $E_7^{(2)}(z_1, z_2)$.

a polynomial order equal to eight has resulted in an MSE of 5.08×10^{-6} and NRMSE of 0.0572%, which is in an acceptable range as mentioned in [36] for the design of 2D filters.

$$\epsilon_{mse} = \frac{1}{4\pi^2} \int_0^{2\pi} \int_0^{2\pi} |Y(\omega_1, \omega_2) - H(\omega_1, \omega_2)|^2 d\omega_1 d\omega_2 \quad (15)$$

$$\epsilon_{nrmse} = \left(\frac{\iint_R |Y(\omega_1, \omega_2) - H(\omega_1, \omega_2)|^2 d\omega_1 d\omega_2}{\iint_R |H(\omega_1, \omega_2, \mu_i)|^2 d\omega_1 d\omega_2} \right)^{\frac{1}{2}} \quad (16)$$

The detailed analysis of the parameters used along with the hardware complexity and error associated with the proposed method, in comparison with that of direct form and Chebyshev structure-based implementation in frequency domain for design example I, are elaborated in Tables II and III. The magnitude response of the 2D filter realized using the proposed technique in comparison with that realized using the

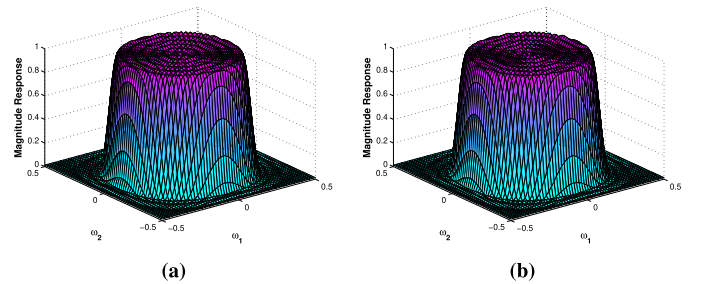


Fig. 5. Magnitude response plots obtained for design example 1 (a) Original filter designed using P_2 transformation, (b) 2D filter response obtained from the proposed structure.

Chebyshev structure used in [21] are shown in Figs. 5a and 5b, respectively.

B. Example II

This example considers the same design example used in [13], where a 2D filter is designed by transforming a 1D

TABLE II

PARAMETERS CHOSEN FOR PROPOSED REALIZATION OF THE 2D FILTER IN EXAMPLE I ALONG WITH THE IMPLEMENTATION COMPLEXITY AND ERROR VALUES OBTAINED

	Example I
1D filter order	30
2D filter size	61×61
Interpolation order (M)	15
Size of polyphase coefficient matrix $P^{(i)}$	14×4
Polynomial order (L)	8
Number of Farrow subfilters	9
Length of each Farrow subfilter	4
Total number of coefficient multipliers in all the Farrow sub-filters	540
Total number of structural multipliers involved in all the Farrow filters	240
Total number of multipliers required in the proposed structure	931
Total number of adders required in the proposed structure	1230
Total number of multipliers required in the direct form implementation	961
Total adders required in the direct form implementation	3600
MSE	5.08×10^{-6}
NRMSE	0.0572%

TABLE III

COMPLEXITY COMPARISONS OF THE PROPOSED 2D FILTER REALIZATION OF DESIGN EXAMPLE I WITH THE REALIZATION OF THE FILTER DESIGNED USING [22] AND [24] IN FREQUENCY DOMAIN

	Proposed method	Direct form structure [24]	Chebyshev based structure [21]
Number of multipliers	931	961	14431
Number of adders	1230	3600	28815

filter of transition width $\Delta f = 0.02\pi$, PB ripple of $\delta_p = 0.02$ and a SB attenuation of $\delta_s = 0.001$, using McClellan transformation. The equivalent 2D transition width in the case of McClellan transformation remains same as $\Delta f = 0.02\pi$. The same 2D filter has been designed using P_2 transformation from a 1D filter of edge frequency equal to 0.45π and $\Delta f = 0.024\pi$, obtained by reverse frequency mapping of the 2D specifications. The equivalent 1D filter obtained is of order 189 and the 2D filter designed using P_2 transformation is of size $(2NP + 1) \times (2NQ + 1) = 379 \times 379$, where $P = Q = 2$. The individual row filters each of length 379, are polyphase decomposed using $M = 63$. The individual coefficients from the different polyphase filters except $E_0(z)$ are polynomial approximated with $L = 28$ for $\mu \in [-0.1, 0.1]$, resulting in a set of 29 Farrow subfilters each of length equal to six. The individual subfilters exhibit symmetric impulse response. The 2D magnitude response obtained using the proposed method is shown in Fig. 6. The complexity comparison of

TABLE IV

COMPLEXITY COMPARISONS OF THE PROPOSED 2D FILTER IN THE DESIGN EXAMPLE II WITH THE STATE OF ART METHODS

	Proposed method	Using sampling kernel interpolation [13]	Direct form realization [24]
Number of multipliers	21735	24740	143641

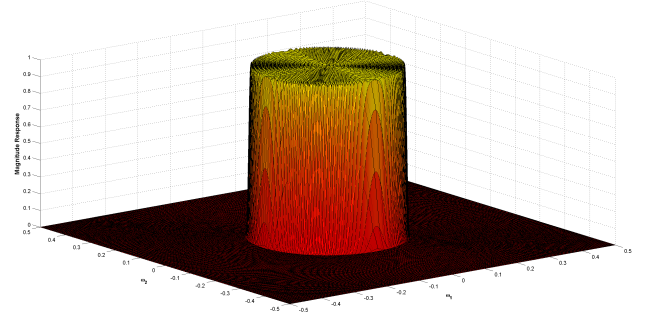


Fig. 6. 3D plot of the magnitude response obtained for the 2D filter in Example II using the proposed polyphase decomposition and Farrow structure approach.

this example has been elaborated in Table IV with respect to the state of art frequency domain realization techniques. Since P_2 transformation produces 2D filters of double the size of 1D filters with $P = Q = 2$, the comparison of the proposed approach with the design method in [13] using $P = Q = 2$ is considered in Table IV.

C. Example III

The design of a fan-type filter of slope angle 25° is considered in this example. The 2D filter is designed by mapping a 1D filter of cut-off frequency $\omega_c = 0.7501\pi$, transition bandwidth $\delta f = 0.01\pi$ and order equal to 60, into a 2D filter of size equal to 61×61 , using the transformation coefficients given in Table I. The 2D filter coefficients obtained are interpolated by row-wise decomposition using $M = 15$ and the coefficients are polynomial approximated using a sixth order polynomial. The resulting 2D response obtained from the original filter designed using McClellan transformation, and that produced from the filter realized from the proposed approach are shown in Figs. 7a and 7b, respectively.

The realization of the fan-type filter using the proposed approach considers the polyphase decomposition of the individual rows of the 2D impulse response using an interpolation order of 15, so that the polyphase terms $E_m(z^{15})$, for $m = 1, 2, \dots, 14$ form the polyphase coefficient matrix $P^{(i)}$ of size equal to 14×4 . The 14 coefficients in each of the columns of $P^{(i)}$ are then polynomial approximated using a polynomial of order equal to six resulting in the Farrow structure.

V. RESULTS AND DISCUSSIONS

The design examples demonstrated in section IV emphasize the effectiveness of the proposed realization structure for time-domain or frequency domain implementation of circular and

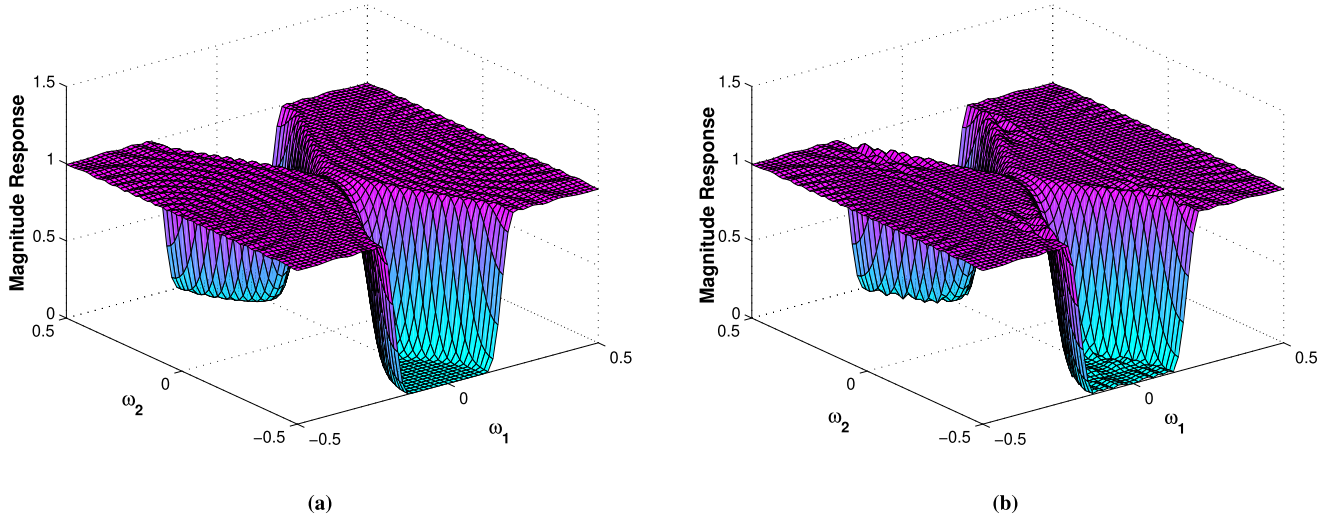


Fig. 7. Magnitude response plots obtained for the fan filters of slope angle 25° (a) Fan filter response of the filter designed using McClellan transformation, (b) Filter response obtained from the proposed method.

fan-type filters. All simulations have been done on an Intel quad core processor using MATLAB (R2010a) software. In Examples I and III, the 2D filter designed is of same size equal to 61×61 . It is observed that quadrantly symmetric frequency response obtained by the transformation methods have the first row and last row consisting of the same coefficients. Hence, the first row, as it can be realized using a single multiplier is exempted from polynomial approximation. Thus, in Example I, 8^{th} order polynomial approximation of the row filters corresponding to the rows starting from the second row to the thirty first row results in individual Farrow structures with 9 symmetric subfilters each of length equal to 4. Thus, only 2 multipliers are required to realize each subfilter and to realize 9 such subfilters, 18 multipliers are required. Also, additionally eight structural multipliers are required in the Farrow structure for realizing the multiplication with the delay parameter μ . Thus, the total number of multipliers to realize the Farrow structure in each row comes out to be 26. The Farrow structure corresponding to each row realizes the polyphase subfilters $E_1(z), \dots, E_{14}(z)$ of the respective row filter. The coefficients of $E_0(z)$ (of length equal to 5) which is not considered in the polynomial approximation requires 5 additional multipliers. Hence, the total number of multipliers required for the realization of each row filter is 31. Thus, the proposed realization requires 931 multipliers including the single multiplier for realizing the first row filter. Similarly, each Farrow structure requires 27 coefficient adders and 8 structural adders accounting to a total of 35 adders. Additionally, the realization of $E_0(z)$ requires 4 more adders resulting in 39 adders for each row starting from the 2^{nd} row to the 31^{st} row (i.e., 30 rows) and 60 adders for realizing the first row filter. Owing to coefficient symmetry of the row filters in the 2D kernel, no additional adders are required for the implementation of last 30 row filters. Thus, the final realization requires the added output of all these 61 row filters, which involves 60 more adders. Thus, the total adders required becomes 1230.

The frequency domain realization of the 2D filter in Example I using the Chebyshev-based structure as in [21]

requires 16 multipliers for the 1D filter coefficients and 15 blocks of $F(\omega_1, \omega_2)$ each of size equal to 61×61 with symmetric coefficients, resulting in a total number of 14431 multipliers. Similarly, each row of $F(\omega_1, \omega_2)$ requires 60 adders and 31 such row implementations require 1860 adders. Additionally, 60 adders are required to compute the added outputs from all the 61 rows, which accounts to a total of 1920 adders for realizing each block of $F(\omega_1, \omega_2)$. Thus, the total number of adders required to realize all the $F(\omega_1, \omega_2)$ blocks is 28,800. Finally, the realization of the 2D filter in frequency domain, based on the structure in [21] requires 28,815 adders. In a similar manner, the implementation complexity in case of Example II is 21735 multipliers. Example III considers the design of a fan-type 2D filter of size equal to 61×61 . The realization of 30 row filters using individual Farrow structures based on polyphase decomposition with $M = 15$ results in a $P^{(i)}$ of size equal to 14×4 (for $i = 2, 3, \dots, 31$). Polynomial approximation of each column of $P^{(i)}$ using a 6^{th} order polynomial results in a Farrow structure with 7 subfilters each of length equal to 4 or order equal to 3. Thus, the total multipliers to realize the Farrow structure in each row comes out to be 20. The Farrow structure corresponding to each row realizes the polyphase subfilters $E_1(z), \dots, E_{14}(z)$ of the respective row filter. The coefficients of $E_0(z)$ (of length equal to 5) which is not considered in the polynomial approximation requires 5 additional multipliers. Hence, the total number of multipliers required for the realization of each row filter is 25. Thus, the realization of 30 such row filters require 750 multipliers. Since the first row filter requires only one multiplier it was not included in the approximation, and thus, the total number of multipliers required becomes 751. In a similar way, the total adders is computed to be 990. On comparing the implementation complexity of the proposed method with the 2D filter obtained in [24], it found that the total number of multipliers and adders required are reduced. The direct form realization of the 2D filter obtained by the design approach in [24] requires 961 multipliers and 3600 adders. A comparison of the implementation complexity of the proposed method and the design method in [24] is

TABLE V

COMPLEXITY COMPARISONS OF THE 2D FAN FILTER IN EXAMPLE III OBTAINED FROM THE PROPOSED METHOD AND THE STATE OF ART METHODS IN [22] AND [24] IN FREQUENCY DOMAIN

	Proposed method	Direct implementation [24]	Chebyshev structure-based [22]
Number of multipliers	751	961	28861
Number of adders	990	3600	108030

given in Table. V. Atleast 21% reduction in the number of multipliers with an NRMSE of 0.2124%, is achieved using the proposed polyphase decomposition based method, compared to the state of art techniques.

VI. CONCLUSION

This paper introduces a low complexity realization technique for 2D filters using polyphase decomposition and polynomial approximation. The proposed method effectively utilizes the quadrantal symmetry of the filter response. The filter coefficients in the individual rows are polyphase decomposed and the polyphase coefficients are polynomial approximated to form the Farrow structure. Thus, the individual rows are realized using separate Farrow structures, by changing the fractional delay parameter, followed by interpolation. The proposed implementation technique requires comparatively reduced hardware complexity than the state of art methods.

REFERENCES

- [1] C.-S. Bouganis, S.-B. Park, G. A. Constantinides, and P. Y. K. Cheung, "Synthesis and optimization of 2D filter designs for heterogeneous FPGAs," *ACM Trans. Reconfigurable Technol. Syst.*, vol. 1, no. 4, pp. 24:1–24:28, Jan. 2009.
- [2] C.-S. Bouganis, G. A. Constantinides, and P. Y. K. Cheung, "A novel 2D filter design methodology for heterogeneous devices," in *Proc. 13th Annu. IEEE Symp. Field-Program. Custom Comput. Machines (FCCM)*, Apr. 2005, pp. 13–22.
- [3] W.-P. Zhu, M. O. Ahmad, and M. N. S. Swamy, "An analytical approach for obtaining a closed-form solution to the least-square design problem of 2-D zero-phase FIR filters," *IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process.*, vol. 41, no. 12, pp. 796–807, Dec. 1994.
- [4] M. O. Ahmad and J.-D. Wang, "An analytical least square solution to the design problem of two-dimensional FIR filters with quadrantly symmetric or antisymmetric frequency response," *IEEE Trans. Circuits Syst.*, vol. 36, no. 7, pp. 968–979, Jul. 1989.
- [5] A. K. Dwivedi, S. Ghosh, and N. D. Londhe, "Low power 2D finite impulse response filter design using modified artificial bee colony algorithm with experimental validation using field-programmable gate array," *IET Sci. Meas. Technol.*, vol. 10, no. 6, pp. 671–678, Aug. 2016.
- [6] R. Matei, "Analytical design methods for directional Gaussian 2D FIR filters," *Multidimensional Syst. Signal Process.*, vol. 29, no. 1, pp. 185–211, 2018.
- [7] S. Dhabal and P. Venkateswaran, "A novel accelerated artificial bee colony algorithm for optimal design of two dimensional FIR filter," *Multidimensional Syst. Signal Process.*, vol. 28, no. 2, pp. 471–493, 2017.
- [8] W. S. Lu, "A unified approach for the design of 2-D digital filters via semidefinite programming," *IEEE Trans. Circuits Syst. I, Fundam. Theory Appl.*, vol. 49, no. 6, pp. 814–826, Jun. 2002.
- [9] A. Aggarwal, M. Kumar, T. K. Rawat, and D. K. Upadhyay, "Optimal design of 2-D FIR digital differentiator using L_1 -norm based cuckoo-search algorithm," *Multidimensional Syst. Signal Process.*, vol. 28, no. 4, pp. 1569–1587, Oct. 2017.

- [10] L. Chen and K. P. Chan, "Design of two-dimensional sharp wideband filters using frequency response masking technique and frequency transformation," *Electron. Lett.*, vol. 45, no. 1, pp. 82–84, Jan. 2009.
- [11] W.-P. Zhu, M. O. Ahmad, and M. N. S. Swamy, "A least-square design approach for 2D FIR filters with arbitrary frequency response," *IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process.*, vol. 46, no. 8, pp. 1027–1034, Aug. 1999.
- [12] T. Williams, M. Ahmadi, and W. C. Miller, "Design of 2D FIR and IIR digital filters with canonical signed digit coefficients using singular value decomposition and genetic algorithms," *Circuits, Syst. Signal Process.*, vol. 26, no. 1, pp. 69–89, 2007.
- [13] K. J. Kim, J. H. Kim, and S. W. Nam, "Design of computationally efficient 2D FIR filters using sampling-kernel-based interpolation and frequency transformation," *Electron. Lett.*, vol. 51, no. 17, pp. 1326–1328, Aug. 2015.
- [14] J. H. McClellan, "The design of two-dimensional digital filters by transformation," in *Proc. 7th Annu. Princeton Conf. Inf. Sci. Syst.*, 1973, pp. 247–251.
- [15] A. Aggarwal, M. Kumar, T. K. Rawat, and D. K. Upadhyay, "Optimal design of 2D FIR filters with quadrantly symmetric properties using fractional derivative constraints," *Circuits, Syst., Signal Process.*, vol. 35, no. 6, pp. 2213–2257, Jun. 2016.
- [16] P. Rajan and M. N. S. Swamy, "Design of circularly symmetric two-dimensional FIR digital filters employing transformations with variable parameters," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-31, no. 3, pp. 637–642, Jun. 1983.
- [17] W. Mecklenbrauker and R. M. Mersereau, "McClellan transformations for two-dimensional digital filtering—Part II: Implementation," *IEEE Trans. Circuits Syst.*, vol. CAS-23, no. 7, pp. 414–422, Jul. 1976.
- [18] T. Saramäki, "Computationally efficient circularly symmetric two-dimensional FIR filters," *IEE Proc. G-Electron. Circuits Syst.*, vol. 131, no. 2, pp. 46–50, Apr. 1984.
- [19] J.-C. Liu and Y.-L. Tai, "Design of 2-D wideband circularly symmetric FIR filters by multiplierless high-order transformation," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 58, no. 4, pp. 746–754, Apr. 2011.
- [20] M. Manuel and E. Elias, "A novel approach for the design of 2D sharp circularly symmetric FIR filters," *Global J. Res. Eng.-Elect. Electron. Eng.*, vol. 12, no. 6, pp. 33–40, May 2012.
- [21] T. Bindima, M. Manuel, and E. Elias, "An efficient transformation for two dimensional circularly symmetric wideband FIR filters," in *Proc. IEEE Region 10 Conf. (TENCON)*, Nov. 2016, pp. 2838–2841.
- [22] T. Bindima, U. K. Shahanas, and E. Elias, "Low complexity fan filters using multiobjective artificial bee colony optimization aided McClellan transformation for directional filtering," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 65-II, no. 12, pp. 2057–2061, Dec. 2018.
- [23] J. H. McClellan and D. Chan, "A 2-D FIR filter structure derived from the Chebyshev recursion," *IEEE Trans. Circuits Syst.*, vol. CAS-24, no. 7, pp. 372–378, Jul. 1977.
- [24] R. Zhao, X. Lai, and Z. Lin, "Weighted least squares design of 2-D FIR filters using a matrix-based generalized conjugate gradient method," *J. Franklin Inst.*, vol. 353, no. 8, pp. 1759–1780, May 2016.
- [25] H. Johansson and O. Gustafsson, "Linear-phase FIR interpolation, decimation, and MTH-band filters utilizing the farrow structure," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 52, no. 10, pp. 2197–2207, Oct. 2005.
- [26] P. G. Patwardhan, B. Patil, and V. M. Gadre, "Polyphase conditions and structures for 2-D quincunx FIR filter banks having quadrantal or diagonal symmetries," *IEEE Trans. Circuits Syst., II, Exp. Briefs*, vol. 54, no. 9, pp. 790–794, Sep. 2007.
- [27] N. Haridas and E. Elias, "Reconfigurable Farrow structure-based FRM filters for wireless communication systems," *Circuits, Syst., Signal Process.*, vol. 36, no. 1, pp. 315–338, 2017.
- [28] C. W. Farrow, "A continuously variable digital delay element," in *Proc. IEEE Int. Symp. Circuits Syst.*, Jun. 1988, pp. 2641–2645.
- [29] C. K. S. Pun, Y. C. Wu, S. C. Chan, and K. L. Ho, "On the design and efficient implementation of the Farrow structure," *IEEE Signal Process. Lett.*, vol. 10, no. 7, pp. 189–192, Jul. 2003.
- [30] J. S. Lim, *Two-Dimensional Signal and Image Processing*. Upper Saddle River, NJ, USA: Prentice-Hall, 1990.
- [31] R. M. Mersereau, W. Mecklenbrauker, and T. Quatieri, "McClellan transformations for two-dimensional digital filtering—Part I: Design," *IEEE Trans. Circuits Syst.*, vol. CAS-23, no. 7, pp. 405–414, Jul. 1976.
- [32] D. Nguyen and M. N. S. Swamy, "Approximation design of 2-D digital filters with elliptical magnitude response of arbitrary orientation," *IEEE Trans. Circuits Syst.*, vol. CAS-33, no. 6, pp. 597–603, Jun. 1986.

- [33] E. Z. Psarakis, V. G. Mertzios, and G. P. Alexiou, "Design of two-dimensional zero phase FIR fan filters via the McClellan transform," *IEEE Trans. Circuits Syst.*, vol. 37, no. 1, pp. 10–16, Jan. 1990.
- [34] E. Z. Psarakis and G. V. Moustakides, "Design of two-dimensional zero-phase FIR filters via the generalized McClellan transform," *IEEE Trans. Circuits Syst.*, vol. 38, no. 11, pp. 1355–1363, Nov. 1991.
- [35] H. K. Kwan and C. L. Chan, "Circularly symmetric two-dimensional multiplierless FIR digital filter design using an enhanced McClellan transformation," *IEE Proc. G Circuits, Devices Syst.*, vol. 136, no. 3, pp. 129–134, Jun. 1989.
- [36] J. J. Shyu, S. C. Pei, and Y. D. Huang, "Design of variable two-dimensional FIR digital filters by McClellan transformation," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 56, no. 3, pp. 574–582, Mar. 2009.



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